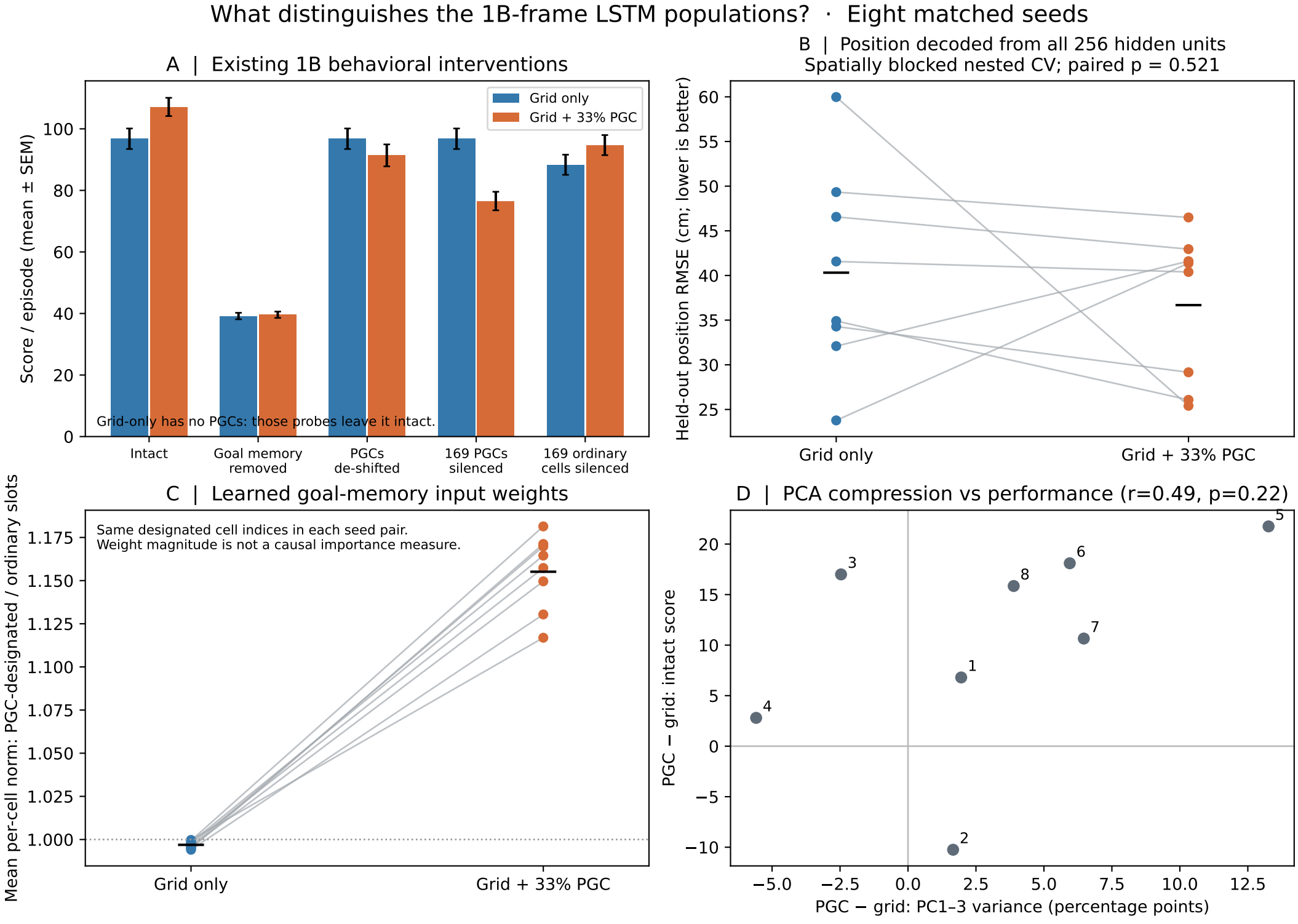
What distinguishes the 1B-frame LSTM populations? · Eight matched seeds
A | Existing 1B behavioral interventions
Score / episode (mean ± SEM)
Grid only
Grid + 33% PGC
Grid-only has no PGCs: those probes leave it intact.
Intact
Goal memory removed
PGCs de-shifted
169 PGCs silenced
169 ordinary cells silenced
B | Position decoded from all 256 hidden units
Spatially blocked nested CV; paired p = 0.521
Held-out position RMSE (cm; lower is better)
Grid only
Grid + 33% PGC
C | Learned goal-memory input weights
Same designated cell indices in each seed pair.
Weight magnitude is not a causal importance measure.
Mean per-cell norm: PGC-designated / ordinary slots
Grid only
Grid + 33% PGC
D | PCA compression vs performance (r=0.49, p=0.22)
PGC − grid: intact score
PGC − grid: PC1–3 variance (percentage points)
Exploratory analysis. Behavior: existing 200-episode evaluations. Hidden states: one controlled eastward replay.